Bilateral monopole saturation and exact bidisk realization

24 September 2026. A continuation of the actual-form and compact-realization arguments. The new feature is the removal of **every topological no-bubble hypothesis**: saturation in both orientations eliminates the complete Weyl curvature on a harmonic-radius blowup. The resulting two global parallel forms give an exact locally symmetric metric on the original smooth manifold. This is a composition theorem, not a new construction of a bidisk quotient or a claim of unrestricted Yau26 realization.

1. Conventions and theorem

Let M be a closed connected oriented smooth four-manifold. Write $M_+ = M$ and $M_- = -M$. A Spin^c structure on M_σ is called an **actual conformal input for** g here if its unperturbed Seiberg–Witten equations are solvable for every metric in $[g]$. The structure is fixed while the positive conformal weight varies. It may depend on the input metric. The two structures below are independent; no common spinor, connection, or equation between the orientations is postulated.

Let a_σ be their real first Chern classes, and set

$$p_+ = \|\operatorname{proj}_{\mathscr{H}^+_g} a_+\|^2_{L^2}, \qquad p_- = \|\operatorname{proj}_{\mathscr{H}^-_g} a_-\|^2_{L^2}. \tag{1}$$

Thus p_- is positive energy, not its negative square in the original orientation. All norms on $W^\pm : \Lambda^\pm \to \Lambda^\pm$ are Hilbert–Schmidt operator norms, norms on two-forms are exterior-algebra norms, and Ricci uses the tensor norm. Define

$$\delta = \|\operatorname{Ric}^\sharp_g + I\|_{\infty,\mathrm{op}}, \qquad r = \fint_M |\operatorname{Ric}_g + g|^2 \leqslant 4\delta^2, \qquad V = \operatorname{Vol}(g), \tag{2}$$

$$\mathfrak{D}_\sigma = \frac{3}{2}\int_M (s^2 + 12|W^\sigma|^2) - 72\pi^2 p_\sigma, \qquad q_\sigma = \mathfrak{D}_\sigma / V \geqslant 0. \tag{3}$$

Nonnegativity is the classical monopole curvature inequality in the respective orientation.

Theorem A (bilateral realization). For every $D, A_0 > 0$ there exists $\varepsilon(D, A_0) > 0$ with the following property. Suppose M, g has actual conformal inputs in both orientations, with

$$p_+ > 0, \quad p_- > 0, \quad \max\{p_+, p_-\} \geqslant A_0, \qquad \operatorname{diam}(g) \leqslant D, \tag{4}$$

and

$$\delta + q_+ + q_- < \varepsilon(D, A_0). \tag{5}$$

Then the **same smooth manifold** admits a smooth metric h with $\operatorname{Ric}_h = -h$ whose universal Riemannian cover is

$$\mathbb{H}^2_{-1} \times \mathbb{H}^2_{-1}. \tag{6}$$

Moreover, h admits global parallel compatible complex structures J_+, J_- inducing the two opposite orientations. In particular it is negative Kähler–Einstein in both orientations.

There is no hypothesis on $\pi_1(M)$, $\pi_2(M)$, an ALE core, the intersection form, or an a priori complex or symplectic structure. No separate volume lower bound, Weyl bound, curvature-energy bound, Sobolev constant, class-square cost, or Ricci-derivative bound is required. The threshold is uniform over varying manifolds and actual class choices.

The conclusion is not that g is already Einstein, nor that M is a globally marked direct product of two surfaces. A compact quotient of the bidisk need not be a direct product. The actual inputs

and smallness of both deficits remain real restrictions. In particular, this theorem is not the earlier one-sided theorem with all conditions simply removed.

If either actual integral Chern class has positive square in its own orientation, its projection has $p_\sigma \geqslant 1$. Then one may set $A_0 = 1$. No negative-square bound on the other class is introduced.

2. The inherited gauge estimates and the volume regime

For each orientation put $\kappa_\sigma = r + q_\sigma$. The actual-form theorem, Sections 2–3, gives, when $\kappa_\sigma \leqslant 1/2880$, a smooth global form $\omega_\sigma \in \Omega^\sigma(M)$ for every $\eta > 0$ sufficiently small, such that

$$\|\omega_\sigma\|_\infty \leqslant 2\sqrt{3}, \qquad \fint (|\omega_\sigma|^2 - 2)^2 \leqslant 720\kappa_\sigma + \eta, \tag{7}$$

$$\fint |\nabla \omega_\sigma|^2 \leqslant 13\kappa_\sigma + 4\sqrt{2\kappa_\sigma} + \eta. \tag{8}$$

These are its zeroth- and first-order conclusions. They have no negative-square cost. The second-order estimate from that theorem is not used. In particular, the proof below does not implicitly add the budget that is needed for a different weak-Bianchi application.

The same classical completed-square estimate, before constructing forms, gives, with $w_\sigma = \sqrt{6}|W^\sigma|$,

$$\fint (w_\sigma - 2)^2 \leqslant q_\sigma + 2r \leqslant 2\kappa_\sigma. \tag{9}$$

Indeed $\int (s + 2w_\sigma)^2 \leqslant 2\mathfrak{D}_\sigma$, whereas $\fint (s+4)^2 \leqslant 4r$, and $w_\sigma - 2 = [(s + 2w_\sigma) - (s+4)]/2$.

Consider a sequence satisfying (4) and $\delta_j + q_{+,j} + q_{-,j} \to 0$. For late terms, $\delta_j \leqslant 1/2$, so $-3g_j \leqslant \mathrm{Ric}_{g_j} \leqslant 3g_j$. Use the scalar monopole inequality in whichever orientation has $p_\sigma \geqslant A_0$:

$$32\pi^2 A_0 \leqslant \int s_-^2 \leqslant 16(1+\delta_j)^2 V_j, \qquad V_j \geqslant v_0 := 8\pi^2 A_0/9. \tag{10}$$

The two choices of orientation give the same scalar curvature and volume. Bishop–Gromov comparison and the diameter bound consequently give

$$V_j \leqslant V_0(D), \qquad c(D, A_0)\rho^4 \leqslant \mathrm{Vol}_{g_j} B_\rho(x) \leqslant C\rho^4 \quad (0 < \rho \leqslant 1). \tag{11}$$

For example $V_0(D) = V_{-1}(D+1)$ suffices; the lower estimate follows by comparing with the entire manifold inside a $(D+1)$-ball.

The elementary inequality $w^2/6 \leqslant (w-2)^2/3 + 4/3$ turns (9) into, for every measurable set $E \subset M_j$,

$$\int_E |W^\sigma_{g_j}|^2 \leqslant \frac{4}{3}\,\mathrm{Vol}_{g_j}(E) + \frac{2}{3}\kappa_{\sigma,j} V_j. \tag{12}$$

This holds in **both** orientations. That is the information absent from the one-sided realization theorem.

3. Topology-free exclusion of harmonic-radius collapse

Lemma 1 (topology-free critical compactness). Fix $K, D, v > 0$ and a modulus $\eta(\rho) \to 0$ as $\rho \downarrow 0$. A family of closed connected smooth four-manifolds satisfying

$$|\mathrm{Ric}| \leqslant K, \qquad \mathrm{diam} \leqslant D, \qquad V \geqslant v, \qquad \sup_x \int_{B_\rho(x)} (|W^+|^2 + |W^-|^2) \leqslant \eta(\rho) \tag{13a}$$

has a uniform positive $C^{1,\alpha}$ harmonic radius for each fixed $\alpha < 1$. It is precompact modulo diffeomorphisms in $C^{1,\beta}$, $\beta < 1$, and weak finite $W^{2,p}$, and has finitely many smooth types. A sequence in this family with $\mathrm{Ric} + g \to 0$ uniformly has a smooth negative Einstein limit on its eventual smooth type and strong finite $W^{2,p}$ convergence. No topological, ALE, or separately supplied curvature-energy hypothesis enters this lemma.

For the sequence in Section 2, (11)–(12) give a common modulus in (13a). Given a target error, choose a tail for which the $\kappa_{\sigma,j}$ terms are small, then choose ρ small for its volume terms; the remaining finitely many smooth metrics also have vanishing small-ball energies. The proof below gives both the lemma and a direct proof for the sequence, without naming that modulus.

Fix $0 < \alpha < 1$ and a standard $C^{1,\alpha}$ harmonic-radius definition with fixed metric-comparability and scaled norm bounds. Suppose the global minimum ρ_j tends to zero, attained at x_j. Rescale

$$\widehat{g}_j = \rho_j^{-2} g_j, \qquad r_h(\widehat{g}_j, x_j) = 1, \qquad r_h(\widehat{g}_j, x) \geqslant 1. \tag{13}$$

The rescaled Ricci norm is $O(\rho_j^2)$, and (11) gives a uniform Euclidean lower growth estimate on every fixed rescaled ball. The rescaled manifolds are complete, their diameters tend to infinity, and the normalized harmonic charts give a pointed complete limit, initially in $C^{1,\beta}$, $\beta < \alpha$.

For clarity, the regularity improvement uses the actual Ricci tensors, not their derivatives. In a harmonic chart the metric equation is

$$g^{ab}\partial_a\partial_b g_{ij} = -2\,\mathrm{Ric}_{ij} + Q_{ij}(g, \partial g). \tag{14}$$

Its limit is weakly Ricci-flat and hence smooth by elliptic bootstrapping. Subtract the equation of this smooth limit from (14). The coefficients converge in C^1, the quadratic lower-order terms converge, and the Ricci tensors converge uniformly to zero. Interior elliptic estimates therefore give strong $W^{2,p}_{\mathrm{loc}}$ convergence for every finite p. Taking $p > 4/(1-\alpha)$ recovers strong $C^{1,\alpha}$ convergence on smaller charts. Curvature consequently converges strongly in local finite L^p spaces. Compatible harmonic charts give the usual embeddings of compact subsets into late terms. These are the same harmonic-coordinate steps spelled out in the compact-realization proof, Section 4.

For any fixed R, four-dimensional scale invariance and (12) give

$$\int_{B_R(x_j, \widehat{g}_j)} |W^{\sigma}_{\widehat{g}_j}|^2 \leqslant \frac{4}{3} C (R\rho_j)^4 + \frac{2}{3} V_0(D)\kappa_{\sigma,j} \longrightarrow 0 \quad (\sigma = +, -). \tag{15}$$

Thus **both** Weyl halves of the Ricci-flat limit vanish. The limit is flat; for the general lemma, (15) is replaced by the bound $\eta(R\rho_j) \to 0$, with exactly the same conclusion. A complete smooth flat four-manifold with positive Euclidean four-dimensional volume growth is $\mathbb{R}^4$: a nontrivial infinite flat deck group reduces the volume-growth degree, whereas a finite group of Euclidean isometries fixes a point and cannot act freely.

On a Euclidean ball of radius larger than two the harmonic-coordinate defining bounds have strict margins. Strong $W^{2,p}$ convergence, followed by solving the harmonic-coordinate Dirichlet problem on a slightly smaller ball, transfers a chart of radius greater than one to x_j. This contradicts (13). Hence the original harmonic radii are bounded below along the sequence.

This argument does not produce a nonflat ALE bubble and then try to classify or forbid its core. It makes every putative blowup **flat before any ALE theorem is invoked**. Neither a global full-curvature energy estimate nor Jiang–Naber, Bando–Kasue–Nakajima, an ALE classification, or a hypothesis on the smooth topology is needed here.

The diameter and harmonic-radius bounds now give a finite-atlas compact limit on a closed smooth manifold. After a subsequence and orientation-preserving diffeomorphisms $\phi_j : N \to M_j$,

$$h_j := \phi_j^* g_j \longrightarrow h \quad \text{in } C^{1,\alpha} \text{ and strong finite } W^{2,p}, \qquad \mathrm{Ric}_h = -h. \tag{16}$$

The same elliptic argument proves smoothness of h; it does not claim smooth convergence of the original approximate metrics. Smooth-type preservation is the ordinary harmonic-atlas compactness conclusion, not a homeomorphism of a possibly singular limit.

4. Two global forms force exact product geometry

Choose $\eta_j \downarrow 0$ and forms from (7)–(8) in both orientations. Pull them back by the **same** diffeomorphisms in (16). Uniform boundedness, the vanishing covariant derivatives in (8), and $h_j \to h$ in C^1 make the two sequences bounded in $H^1(N,h)$. After a common subsequence they converge strongly in L^2 to global forms ω_+, ω_-. The connection differences tend uniformly to zero; hence their distributional covariant derivatives for h vanish. Elliptic regularity and the amplitude estimates give

$$\nabla^h \omega_\pm = 0, \qquad *_h \omega_\pm = \pm \omega_\pm, \qquad |\omega_\pm|_h^2 = 2. \tag{17}$$

In particular, neither form vanishes. There is no eigenline sign choice and no two-fold cover: the approximating forms were global.

Define $J_\pm$ by $h(J_\pm X, Y) = \omega_\pm(X,Y)$. In four dimensions (17) gives $J_\pm^2 = -I$. Self-dual and anti-self-dual skew endomorphisms commute, so

$$K = -J_+ J_-, \qquad K^* = K, \qquad K^2 = I, \qquad \mathrm{tr}\, K = 0, \qquad \nabla K = 0. \tag{18}$$

The trace follows equally from the orthogonality of the two forms. Consequently $TN = E_1 \oplus E_2$, the two orthogonal rank-two parallel eigenbundles of K. Parallel distributions are integrable and their leaves are totally geodesic. On the complete simply connected universal cover the de Rham splitting is a global Riemannian product of two complete simply connected surfaces. Since $\mathrm{Ric}_h = -h$, the Ricci tensor of each factor is minus its metric. In dimension two this means constant Gaussian curvature -1. Both factors are $\mathbb{H}^2_{-1}$, proving (6). Parallelness of $J_\pm$ also proves their integrability and the global Kähler assertions directly.

If Theorem A failed for fixed D, A_0, choose counterexamples with $\delta_j + q_{+,j} + q_{-,j} < 1/j$. Sections 2–4 give (16)–(18). Pulling h, J_+, J_- forward to each late M_j contradicts its being a counterexample. This proves a fixed threshold uniform across changing smooth types. It does not assume that an individual small-error metric can be improved into a whole sequence on its own manifold. □

5. Raw Ricci-energy formulation

Let

$$T_\pm = 2\chi(M) \pm 3\tau(M), \quad R_2 = \int_M |\mathrm{Ric}|^2, \quad S_2 = \int_M s^2, \quad P = \int_M |\mathrm{Ric}^\circ|^2.$$

Since $R_2 = P + S_2/4$ and $4\pi^2 T_\sigma = S_2/24 + 2\int |W^\sigma|^2 - P/2$,

$$\boxed{\mathfrak{D}_\sigma = \frac{9}{2}\big[R_2 - 8\pi^2(2p_\sigma - T_\sigma)\big].} \tag{19}$$

This is a classical Chern–Weil identity, the exact deficit form of LeBrun's Ricci-energy estimate. It is not a new inequality.

Suppose independent actual projection budgets are available, $p_\sigma \geqslant A_\sigma > 0$, and $\max(A_+, A_-) \geqslant A_0$. Define the nonnegative averaged two-sided energy excess

$$e_{\mathrm{bi}} = \frac{R_2 - 8\pi^2(A_+ + A_- - 2\chi)}{V}. \tag{20}$$

Adding (19) and using $T_+ + T_- = 4\chi$ gives the exact decomposition

$$\boxed{e_{\mathrm{bi}} = \frac{q_+ + q_-}{9} + \frac{8\pi^2}{V}\Big[(p_+ - A_+) + (p_- - A_-)\Big].} \tag{21}$$

Every term on the right is nonnegative. Thus no cancellation between the orientations is hidden by the single energy condition.

Corollary B (energy-to-geometry realization). Under the actual-input, budget and diameter assumptions just stated, without any topological restriction, the condition

$$\delta + 9e_{\mathrm{bi}} < \varepsilon(D, A_0) \tag{22}$$

implies the full conclusion of Theorem A.

Indeed (21) gives $q_+ + q_- \leqslant 9e_{\mathrm{bi}}$, so (5) follows. This replaces the two separately displayed functional deficits by a raw Ricci-energy excess above the average of two genuine gauge bounds. It does not replace the existence of the two actual inputs by numerical cohomology data, and it does not claim that (20) is the attained global Ricci-energy infimum on every supplied topology.

The exact counterpart using the actual projections is

$$q_+ + q_- = 9\,\frac{R_2 - 8\pi^2(p_+ + p_- - 2\chi)}{V}. \tag{23}$$

Neither (22) nor (23) follows from $\delta \ll 1$ alone. Their purpose is a checkable interface for independent topological or variational inputs.

6. Consequences, tests, and scope

6.1 Topology is now a conclusion, not an assumption

Let $\mathscr{B}$ denote the smooth manifolds admitting a metric with universal Riemannian cover (6) and the two global parallel structures of Theorem A. If $M \notin \mathscr{B}$, then every input meeting (4) obeys

$$\delta + q_+ + q_- \geqslant \varepsilon(D, A_0). \tag{24}$$

Examples of independently checkable exclusions from $\mathscr{B}$ are $\pi_2(M) \neq 0$, torsion in $\pi_1(M)$, or $\tau(M) \neq 0$. Indeed (6) is contractible; a finite-order deck isometry of this Hadamard space has a fixed point and cannot act freely; and for the product metric both Weyl halves have the same squared norm 2/3, so the signature is zero. The output also has $\chi(M) > 0$.

Thus the theorem does not merely improve a threshold within the old no-ALE-core class. It tests inputs on **arbitrary** smooth four-manifolds and, when they pass, deduces the topology and geometry that the earlier realization theorem had partly assumed. The new condition is bilateral saturation; neither theorem subsumes the other on all its inputs.

Equation (24) is a joint deficit gap, not a pointwise-only gap on every non-bidisk manifold. No claim follows that such a manifold has no other Einstein metric.

6.2 Actual equality models

For $M = \Sigma_g \times \Sigma_h$, $g, h \geqslant 2$, take the product of curvature -1 metrics and complex structures $J_+ = (J_1, J_2)$, $J_- = (J_1, -J_2)$. The canonical Spin^c structures in the two orientations supply actual classes: both are general-type Kähler surfaces with $b_\pm > 1$, and their canonical Seiberg–Witten invariants are nonzero. Set $A = 8(g-1)(h-1) = 2\chi(M)$. Then

$$V = 2\pi^2 A, \quad p_+ = p_- = A, \quad |W^+|^2 = |W^-|^2 = 2/3, \quad \delta = q_+ = q_- = e_{\mathrm{bi}} = 0 \tag{25}$$

with $A_+ = A_- = A$. This verifies simultaneous saturation in a real nonempty family. It is a classical equality model, not a newly constructed family. The theorem's increment is the uniform implication for arbitrary smooth inputs and changing topologies, without a presumed product structure.

Unequally scaled hyperbolic products remain cscK in both orientations and have $q_+ = q_- = 0$, while generally $\delta \neq 0$. They test why simultaneous functional equality must not be identified with the input metric itself being Einstein. Their underlying manifold already admits the normalized product metric, consistently with the existence conclusion.

6.3 Diameter cannot simply be dropped, even with both inputs

The fixed nonzero-shear self-gluings of Round 11, Section 6, carry symplectic structures in **both** orientations. Their canonical classes are actual monopole inputs with $p_\pm \geqslant A = 2\chi > 0$; this follows from $b_\pm > 1$ and Taubes as recorded in the variational proof, Sections 1–2. The actual normalized metrics g_ℓ satisfy

$$\tau = 0, \quad V_\ell = 2\pi^2 A, \quad \delta_\ell \to 0, \tag{26}$$

and the global topology proof excludes a Kähler structure on every finite cover.

Using (19), $R_2/V \leqslant 4(1+\delta_\ell)^2$, $T_\pm = A$, and $p_\pm \geqslant A$, one obtains the simultaneous bounds

$$0 \leqslant q_{\pm,\ell} \leqslant 18\Big[(1+\delta_\ell)^2 - 1\Big] = 36\delta_\ell + 18\delta_\ell^2 \longrightarrow 0. \tag{27}$$

Thus bilateral actual saturation and pointwise almost-Einstein input do occur on a fixed manifold with no bidisk metric, when diameter is unrestricted. Theorem A forces $\mathrm{diam}(g_\ell) \to \infty$. Equation (27) uses the direct Ricci-energy estimate and is slightly tighter than the separate scalar/traceless estimates in the one-sided note; that numerical improvement is not a separate mathematical achievement.

These are inherited constructions and a new consequence, not a new counterexample construction. They disprove diameter-free **bidisk or KE** realization with these inputs. They do not decide whether those manifolds carry a general non-Kähler Einstein metric, and therefore do not furnish Yau26 counterexamples.

6.4 What is and is not new in the proof

The scalar and curvature-functional bounds, Chern–Weil algebra, harmonic compactness and de Rham splitting are classical. The actual global form estimates are inherited first-line results. The new composition uses the **second actual orientation** twice: to rule out all harmonic-radius blowups without topology, and to upgrade the limiting KE structure to an exact bidisk realization. It thereby connects two-sided near-optimal Ricci energy to exact geometry on the same smooth manifold.

The result establishes the Yau26 existence output on its stated inputs; locally symmetric negative KE is more than the required output. The remaining limitations are the diameter, actual bilateral class supply, and simultaneous-saturation inputs. The proof supplies neither a dimension-only threshold nor an effective numerical constant. No exhaustive priority claim is made.

7. Dependencies and finite verification

- Inherited actual-form theorem, Sections 2–3: global forms and the explicit amplitude/gradient estimates.
- One-sided realization, Sections 3–6: the original composition and one-form limit. The present argument replaces its topological no-bubbling step, not its form supply.
- Harmonic-coordinate realization proof, Sections 3–5: the metric-equation convergence and preservation of smooth type. Its ALE and classification steps are not used here.
- LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Theorem 2.3 and Propositions 4.1–4.2: actual-conformal curvature inequality, cscK equality, and Ricci-energy identity. The energy reformulation above is classical algebra; it is not attributed as a new part of this continuation.
- Anderson, *Convergence and rigidity of manifolds under Ricci curvature bounds*, and Zhang–Zhu, *Bounds on harmonic radius and limits of manifolds with bounded Bakry–Émery Ricci curvature*, Section 2, especially its construction of larger harmonic charts in a Euclidean limit: the harmonic-coordinate compactness framework. The Ricci-flat equation and strong finite-Sobolev upgrade needed here are written explicitly in Section 3.
- Finite check verifies (19), (21), the product normalization, and the commuting opposite-handed complex structures. It is not a computer proof of gauge solvability, harmonic compactness, or the realization theorem.